Adaptive Spectral Thresholding via Kurtosis-Guided Squared Envelope Analysis for Load-Invariant Bearing Fault Detection

Member 1, Member 2, Member 3, and Member 4

***Abstract*—Rolling element bearing diagnostics rely on envelope spectrum analysis to extract the periodic impulse rates produced by localized surface defects. In industrial operation, however, variable mechanical load alters the shaft slip, the structural damping and the broadband noise floor, so that fixed bandpass filters and static spectral thresholds exhibit high false-alarm rates under heavy load and missed detections under light load. This paper presents a training-free pure Digital Signal Processing (DSP) framework that combines Kurtosis-guided resonance demodulation with local noise-floor adaptive thresholding in the Squared Envelope Spectrum (SES). Cascade second-order-section (SOS) zero-phase IIR filtering, discrete Hilbert transform demodulation, Hann-windowed Fast Fourier Transform (FFT) and order-statistic Median–Median Absolute Deviation (Median–MAD) Constant False Alarm Rate (CFAR) thresholding allow the detector to track load-induced noise-floor shifts and shaft slip automatically. The processing chain is mathematically interpretable, requires no training data, and is sized for fixed-point embedded implementation; it is applied here to the multi-load CWRU and MFPT bearing benchmarks.**

***Index Terms*—Bearing fault diagnosis, Squared Envelope Spectrum (SES), Hilbert transform, spectral kurtosis, IIR/FIR filter design, CFAR, Median–MAD adaptive thresholding.**

I. Introduction

ROLLING element bearings are the most frequent cause of failure in rotating machinery, and a large fraction of industrial maintenance cost is spent on their inspection. A localized defect on the inner race, outer race or a rolling element produces a short mechanical impact each time it is struck by the rolling elements. Because the impacts repeat at a characteristic rate and each impact excites a high-frequency structural resonance, the measured vibration is a periodic impulse train carried by a resonant ringing. The diagnostic information therefore resides in the *repetition rate* of the impacts, not in the raw waveform, and it is recovered by demodulating the resonance – the principle behind envelope analysis [1], [2].

Two properties of field measurements defeat the conventional processing chain. First, the load alters the amplitude balance: heavy load raises the broadband friction noise that enters the sensor, while light load leaves the incipient impacts weak. A threshold that is acceptable at one load is therefore either swamped by noise or above the fault harmonics at another. Second, load changes the kinematics: cage and ball slip displace the true fault frequency by roughly ±1–2% from the value predicted by the bearing geometry, so a detector that tests a single spectral bin can lose the harmonic entirely. Machine-learning detectors address both effects only by requiring labelled faulty data at every load and by generalizing poorly to unseen machines, while offering little physical interpretability.

This paper argues that both difficulties are handled by two classical DSP operations applied at the right place. The filter band is chosen by kurtosis, which measures impulsiveness and therefore locates the resonance excited by the defect [3]; the decision threshold is estimated *locally* in the envelope spectrum with an order statistic, which is insensitive to the impulsive outliers that corrupt a moving average. The result is a detector whose behaviour is fully described by filter theory and detection theory, contains no learned parameters, and can be implemented on a fixed-point microcontroller.

The contributions of this paper are:

- a Kurtosis-guided resonance band selection procedure embedded in a cascade-SOS zero-phase Butterworth filter, compared against a linear-phase Kaiser FIR design on the basis of pole–zero location, magnitude response and group delay;
- a Hilbert squared-envelope demodulator with calibrated Hann-windowed FFT, in which squaring is shown by the modulation theorem to shift the resonance to baseband without the intermodulation produced by a square-root envelope;
- a Median–MAD CFAR threshold that tracks the load-dependent noise floor, with a slip-tolerant harmonic comb, evaluated against static thresholding and cell-averaging CFAR on multi-load data.

II. Signal Model and Theoretical DSP Framework

A. Continuous-to-Discrete Fault Signal Model

Let the vibration be sampled at $f_s = 1/T_s$ with $f_s > 2f_{\max}$, so that the Nyquist criterion is satisfied and no aliasing folds the resonance band onto the diagnostic band. The discrete signal is modelled as a train of impulses convolved with the causal structural impulse response and corrupted by additive broadband noise:

$$x[n] = \sum_{m=-\infty}^{\infty} A_m \, h[n - mN_p - \tau_m] + w[n], \tag{1}$$

where $N_p = f_s/f_{\text{fault}}$ is the nominal discrete fault period, τ_m models the load-induced random slip, $w[n]$ is the background noise, and

$$h[n] = e^{-\sigma n T_s} \cos(\omega_r n)\, u[n] \tag{2}$$

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is a causal, BIBO-stable underdamped resonance. In the z-plane, $h[n]$ corresponds to a complex-conjugate pole pair

$$z_{1,2} = re^{\pm j\omega_r}, \qquad r = e^{-\sigma T_s} < 1, \tag{3}$$

so stability follows from $|z_{1,2}| < 1$. Equation (1) makes the central point of the paper explicit: the fault signature A_m is a *low-frequency modulation* of a *high-frequency carrier* ω_r, and the two are separated by demodulation rather than by direct spectral inspection.

B. Pre-Processing and Kurtosis-Guided Bandpass Filter Design

1) Mean removal. The sensor signal contains a DC offset from the accelerometer bias and the static gravity component. Subtracting the sample mean

$$x_0[n] = x[n] - \frac{1}{N}\sum_{m=0}^{N-1} x[m] \tag{4}$$

prevents the zero-frequency bin from dominating the spectrum and from leaking into the low-frequency region where the fault harmonics lie.

2) Sub-band kurtosis selection. The resonance frequency ω_r is not known a priori. A filter bank of B bands of width $\Delta\omega$ spanning $[0, f_s/2]$ is applied to $x_0[n]$, and the normalized fourth-order moment

$$K_y = \frac{\mathbb{E}\{(y-\mu_y)^4\}}{\left(\mathbb{E}\{(y-\mu_y)^2\}\right)^2} \tag{5}$$

is evaluated at the output of each band. A Gaussian noise band yields $K_y \approx 3$; a band containing the impulse-excited resonance yields $K_y > 3$ because the impacts are rare and large. The band with maximum K_y defines $[\omega_{p1}, \omega_{p2}]$ and hence the demodulation carrier.

3) Filter design. Two designs are compared. The linear-phase Type-I Kaiser-windowed FIR filter of length L has

$$h_{\mathrm{FIR}}[n] = h_{\mathrm{ideal}}[n]\, w_{\mathrm{Kaiser}}[n], \quad w_{\mathrm{Kaiser}}[n] = \frac{I_0\left(\beta\sqrt{1-\tilde{n}^2}\right)}{I_0(\beta)}, \tag{6}$$

where $\tilde{n} = 2n/(L-1) - 1$ is a normalized index and the shape parameter β trades transition width against stopband attenuation. The alternative is a 4th-order Butterworth bandpass, obtained by the bilinear transform [4]

$$s = \frac{2}{T_s}\,\frac{1-z^{-1}}{1+z^{-1}}, \tag{7}$$

with the analog band edges pre-warped according to $\Omega = \frac{2}{T_s}\tan(\omega/2)$ so that the digital cutoff lands exactly on the specified frequency.

4) Cascade SOS realisation. A high-order transfer function evaluated in direct form suffers from coefficient quantization: small perturbations of the denominator coefficients move the poles of (3), and for narrow bands the filter can even become unstable in finite precision. The design is therefore factored into second-order sections,

$$H(z) = g\prod_{i=1}^{K}\frac{b_{0i} + b_{1i}z^{-1} + b_{2i}z^{-2}}{1 + a_{1i}z^{-1} + a_{2i}z^{-2}}, \tag{8}$$

each of which keeps its own pole pair well conditioned.

5) Zero-phase filtering. Because the diagnostic decision depends on the *time* of each impulse, the filter must not disperse the transients. Forward–backward filtering applies $H(z)$ and then $H(z^{-1})$,

$$Y_{\mathrm{BP}}(z) = H(z)H(z^{-1})X(z) \implies \left|H(e^{j\omega})\right|^2 e^{-j0}, \tag{9}$$

which squares the magnitude response and cancels the phase, so the group delay is exactly zero and the impulse positions are preserved.

C. Hilbert Demodulation and Windowed Squared Envelope Spectrum

After the Kurtosis-guided zero-phase bandpass stage, $y_{\mathrm{BP}}[n]$ holds short bursts of structural ringing that repeat at the fault rate f_{fault}. The FFT of $y_{\mathrm{BP}}[n]$ concentrates its energy around the resonance ω_r and shows almost nothing at f_{fault}, so the repetition rate has to be demodulated before it can be measured at all. The analytic signal is built with the Discrete Hilbert Transform [5]:

$$z[n] = y_{\mathrm{BP}}[n] + j\hat{y}_{\mathrm{BP}}[n], \tag{10}$$

where $\hat{y}_{\mathrm{BP}}[n] = \mathcal{H}\{y_{\mathrm{BP}}[n]\}$ is the transform of the filtered signal. The transformer is an ideal 90° phase shifter,

$$H_{\mathrm{Hilbert}}(e^{j\omega}) = -j\ \mathrm{sgn}(\omega), \tag{11}$$

which moves every positive-frequency component by $-\pi/2$ and removes the negative frequencies.

A conventional envelope takes the square root of $y_{\mathrm{BP}}^2 + \hat{y}_{\mathrm{BP}}^2$. The squared envelope is used instead [1]:

$$e[n] = |z[n]|^2 = z[n]z^*[n] = y_{\mathrm{BP}}^2[n] + \hat{y}_{\mathrm{BP}}^2[n]. \tag{12}$$

Two consequences follow from LTI theory. Squaring in time is auto-convolution in frequency, so the DTFT modulation theorem translates the carrier ω_r to baseband (0 Hz) as a clean copy of itself. The square root offers no such guarantee: being non-linear, it scatters spurious intermodulation harmonics that can bury a weak incipient fault. Squaring also weights the impacts. A large peak grows quadratically while the background does not, so the fault-to-noise ratio rises before any threshold is applied.

Since $e[n] \geq 0$ everywhere, the squared envelope carries a large DC term. Its mainlobe would leak across the low-frequency bins and hide the cage and shaft harmonics, so the mean is removed first:

$$e_0[n] = e[n] - \mu_e, \qquad \mu_e = \frac{1}{N}\sum_{n=0}^{N-1} e[n]. \tag{13}$$

Fault frequencies rarely land on a DFT bin centre, and 107.36 Hz with $\Delta f = f_s/N_{\mathrm{FFT}}$ is no exception. Rectangular truncation then leaks sidelobes into the neighbouring bins, which an N-point Hann window [6] suppresses:

$$w[n] = \tfrac{1}{2}\left(1 - \cos\frac{2\pi n}{N-1}\right), \qquad n = 0, 1, \ldots, N-1. \tag{14}$$

Tapering the ends to zero scales the coherent amplitude by one half, and the coherent gain

$$CG = \frac{1}{N}\sum_{n=0}^{N-1} w[n] = \tfrac{1}{2} \tag{15}$$

restores the amplitude scale. An N_{FFT}-point radix-2 FFT finally gives the one-sided Squared Envelope Spectrum

$$\mathrm{SES}[k] = \frac{2}{N \cdot CG}\left|\sum_{n=0}^{N-1} w[n]\, e_0[n]\, e^{-j\frac{2\pi}{N_{\mathrm{FFT}}}kn}\right|, \tag{16}$$

for $k = 0, 1, \ldots, N_{\mathrm{FFT}}/2$, where bin k sits at $f_k = k\, f_s/N_{\mathrm{FFT}}$ Hz.

D. Local Noise-Floor Adaptive Spectral Thresholding

Radial and axial load shift the SES floor in both directions. At 0 HP the contact force between the rolling elements and the raceways is low, the impact modulation is weak, and a fixed threshold set for normal operation sits above the fault peaks and misses them, a false negative. At 3 HP friction lifts the whole floor, and the same threshold is crossed by noise in dozens of bins, a false alarm.

An automated sliding-window Constant False Alarm Rate (CFAR) rule [7] covers both cases without training data and stays load-invariant by construction. For a test bin k, the N_G bins on either side are excluded from the noise estimate. The Hann mainlobe is four bins wide, so $N_G = 4$ keeps the skirts of a fault peak out of its own estimate. Beyond the guard band, $N_R = 25$ bins per side form the reference window

$$\begin{aligned}\mathcal{W}_k = \big\{\mathrm{SES}[m] \;\big|\; & m \in [k - N_G - N_R,\, k - N_G - 1] \\ & \cup\, [k + N_G + 1,\, k + N_G + N_R]\big\}.\end{aligned} \tag{17}$$

Two estimators are compared over $\mathcal{W}_k$.

1) Cell-Averaging CFAR: Cell averaging treats the window as a moving-average filter:

$$\hat{\mu}[k] = \frac{1}{2N_R}\sum_{x\in\mathcal{W}_k} x, \qquad T_{\mathrm{CFAR}}[k] = \alpha\,\hat{\mu}[k], \tag{18}$$

with the multiplier $\alpha = 3.5$. It follows the broadband floor, but bearing spectra defeat it through harmonic target masking. A tall shaft harmonic (1×, 2× RPM) or a neighbouring fault harmonic lands inside $\mathcal{W}_k$, the mean moves with that outlier, and T_{CFAR} rises right beside the fault it should expose.

2) Proposed Order-Statistic Median–MAD Threshold: The proposed rule keeps the order statistics and drops the mean:

$$\begin{aligned}\hat{m}[k] &= \mathrm{median}\,(\mathcal{W}_k)\,, \\ \mathrm{MAD}[k] &= \mathrm{median}\,(|\mathcal{W}_k - \hat{m}[k]|)\,.\end{aligned} \tag{19}$$

For Gaussian noise the MAD rescales to a robust standard deviation through the consistency factor 1.4826 [8],

$$\hat{\sigma}_{\mathrm{local}}[k] = 1.4826\,\mathrm{MAD}[k], \tag{20}$$

and the threshold becomes

$$T_{\mathrm{MAD}}[k] = \hat{m}[k] + \gamma\,(1.4826\,\mathrm{MAD}[k])\,, \tag{21}$$

where $\gamma = 4.5$ sets the margin in local standard deviations above the median. The breakdown point of the median is 50%, so half the window can be contaminated by shaft or fault harmonics before T_{MAD} moves. The threshold tracks the true floor smoothly from 0 HP to 3 HP.

TABLE I
CWRU Drive-End Bearing Characteristic Frequencies

Load	f_r (Hz)	BPFO	BPFI (Hz)	BSF	FTF
0 HP (1797 rpm)	29.95	107.36	162.18	70.58	11.93
1 HP (1772 rpm)	29.53	105.86	159.92	69.60	11.76
2 HP (1750 rpm)	29.17	104.56	157.94	68.74	11.62
3 HP (1730 rpm)	28.83	103.36	156.14	67.95	11.48

3) Slip-Tolerant Harmonic Comb: Slip remains. Heavy radial load makes the rolling elements skid by ±1–2% [1], and the measured fault frequency drifts from the kinematic value. The comb therefore tests a band rather than a single bin,

$$\mathcal{B}_h = \left[h f_{\mathrm{fault}}(1-\delta_{\mathrm{slip}}),\; h f_{\mathrm{fault}}(1+\delta_{\mathrm{slip}})\right], \tag{22}$$

for every $f_{\mathrm{fault}} \in \{f_{\mathrm{BPFO}}, f_{\mathrm{BPFI}}, f_{\mathrm{BSF}}\}$ and $h \in \{1, 2, 3\}$, with $\delta_{\mathrm{slip}} = 1.5\%$. Harmonic h is declared present when the largest spectral excess inside the band is positive,

$$\max_{f_k\in\mathcal{B}_h}\,(\mathrm{SES}[k] - T_{\mathrm{MAD}}[k]) > 0. \tag{23}$$

III. Experimental Setup and Datasets

A. Bearing Kinematics

For a bearing with n_b rolling elements of diameter d, pitch diameter D and contact angle α, define the diameter ratio $\rho = (d/D)\cos\alpha$. The theoretical fault frequencies at shaft rate f_r are then

$$\begin{aligned} f_{\mathrm{BPFO}} &= \frac{n_b}{2} f_r\,(1-\rho)\,, \qquad f_{\mathrm{BPFI}} = \frac{n_b}{2} f_r\,(1+\rho)\,, \\ f_{\mathrm{BSF}} &= \frac{D}{2d} f_r\left(1-\rho^2\right), \qquad f_{\mathrm{FTF}} = \frac{f_r}{2}\,(1-\rho)\,. \end{aligned} \tag{24}$$

Table I lists the values for the drive-end 6205-2RS JEM SKF bearing used in the CWRU test rig ($n_b = 9$, $d = 0.3126$ in, $D = 1.537$ in, $\alpha = 0°$) at the four motor loads.

B. Datasets

The framework is evaluated on two public benchmarks. *CWRU:* the Case Western Reserve University Bearing Data Center records accelerometer signals at $f_s = 12$ kHz and 48 kHz for motor loads of 0, 1, 2 and 3 HP, with artificially seeded outer-race, inner-race and ball defects of 7, 14 and 21 mil [9]. *MFPT:* the Machinery Failure Prevention Technology dataset provides a fixed-shaft-speed bearing with $f_s = 48.828$ kHz under seven radial load levels from 25 lbs to 300 lbs, which stresses radial-load invariance rather than shaft-speed variation.

C. Pipeline Parameters

Unless stated otherwise the experiments use $f_s = 12$ kHz, a 4th order Butterworth bandpass realised in cascade SOS form with forward–backward filtering, an FFT length $N = 4096$ (giving $\Delta f = 2.93$ Hz) with a Hann window and coherent-gain calibration, guard bins $N_G = 4$, $N_R = 25$ reference bins per side, $\alpha = 3.5$ for CA-CFAR and $\gamma = 4.5$ for the Median–MAD rule. The static reference threshold is calibrated once

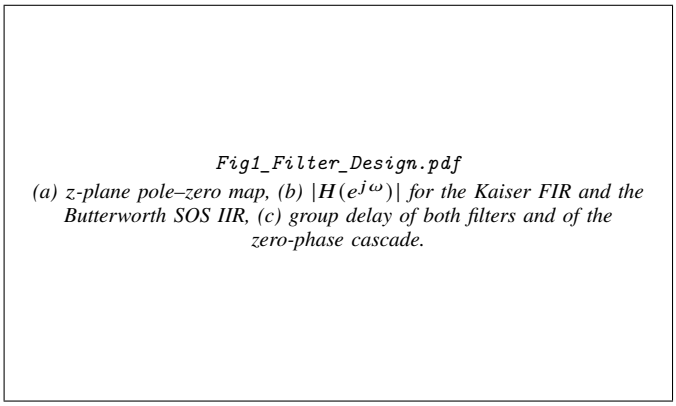


Fig. 1. Filter design: pole–zero map, magnitude response and group delay of the Kaiser FIR and the 4th-order Butterworth SOS bandpass.

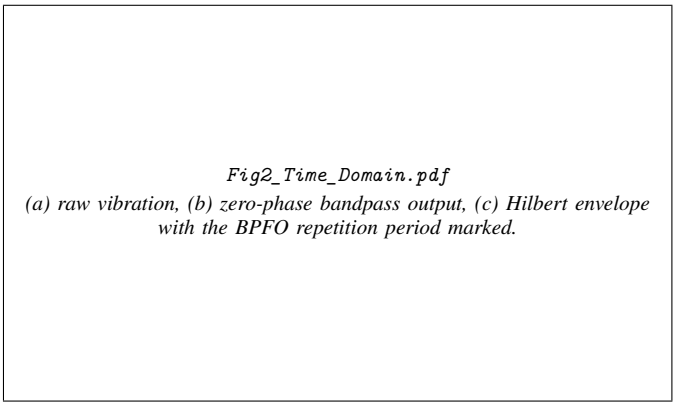


Fig. 2. Time-domain progression from the raw record to the demodulated envelope.

on the 0 HP record at a 1% false-alarm rate and then held fixed, so that any change in its performance across loads is attributable to the load itself.

IV. Results and Discussion

A. Filter Design and Pole–Zero Analysis

Fig. 1 compares the two candidate filters. The pole–zero map confirms that the SOS denominator places a conjugate pair just inside the unit circle, at radius $r = e^{-\sigma T_s}$ set by the selected band, while the zeros sit on the unit circle at the band edges: the stopband is therefore exact at DC and at Nyquist and the passband is stable by construction. The magnitude responses of the Kaiser FIR and the Butterworth IIR are comparable within the selected band, but differ in the transition region, where the FIR length sets the slope. The group-delay panel is the decisive comparison: the linear-phase FIR delays all frequencies equally, the causal IIR delays them unequally, and forward–backward filtering removes both, which is the reason the zero-phase SOS structure is used for impulse preservation.

B. Time-Domain Progression

Fig. 2 shows the three stages of the pipeline on a short segment. The raw record is dominated by low-frequency shaft components and broadband noise, in which the impacts are not visually separable. After the Kurtosis-guided zero-phase bandpass the resonance ringing of each impact becomes visible as a burst, and after the Hilbert transform the envelope exposes the repetition period directly: the spacing between successive maxima equals $1/f_{\text{BPFO}}$, which is the quantity the SES resolves in the frequency domain.

C. Multi-Load Adaptive Thresholding

Fig. 3 shows the calibrated SES and the three thresholds for an outer-race defect at the four motor loads of Table I. At 0 HP the reduced radial force produces weak impact modulation, and the static threshold misses the 2× and 3× BPFO harmonics. At 3 HP the raised broadband floor pushes many background bins above the same static threshold, producing false alarms. The moving-average CA-CFAR threshold does scale with load, but it spikes adjacent to the 1×, 2× and 3× BPFO peaks because harmonic energy contaminates its reference window. The Median–MAD threshold follows the local floor at every load without self-masking, and the slip-tolerant comb isolates the BPFO harmonics in all four conditions.

TABLE II
Thresholding Benchmark on the Kurtosis-Guided SES

Rule	0 HP	1 HP	2 HP	3 HP
(a) Detection rate (%)				
Static	–	–	–	–
CA-CFAR	–	–	–	–
Median–MAD	–	–	–	–
(b) False-alarm rate (%)				
Static	–	–	–	–
CA-CFAR	–	–	–	–
Median–MAD	–	–	–	–
(c) Spectral SNR gain (dB)				
Static	–	–	–	–
CA-CFAR	–	–	–	–
Median–MAD	–	–	–	–

D. Quantitative Benchmark

Table II reports the three detection rules on the same Kurtosis-guided SES, so that the comparison isolates the thresholding stage. Spectral SNR gain is defined as the ratio of the mean comb peak to the median out-of-comb floor in dB, the detection rate is the fraction of segments in which all three harmonics are declared present, and the false-alarm rate is the fraction of out-of-comb bins that exceed the threshold. The static rule is expected to trade false alarms at heavy load against missed detections at light load, CA-CFAR to lose detections through target masking, and the proposed rule to hold a near-constant false-alarm rate across the load sweep.

V. Conclusion and Hardware Feasibility

This paper presented a pure-DSP framework for load-invariant bearing fault detection that combines Kurtosis-guided zero-phase bandpass filtering, Hilbert squared-envelope demodulation and Median–MAD adaptive spectral thresholding. Filter design, demodulation and detection are each traceable

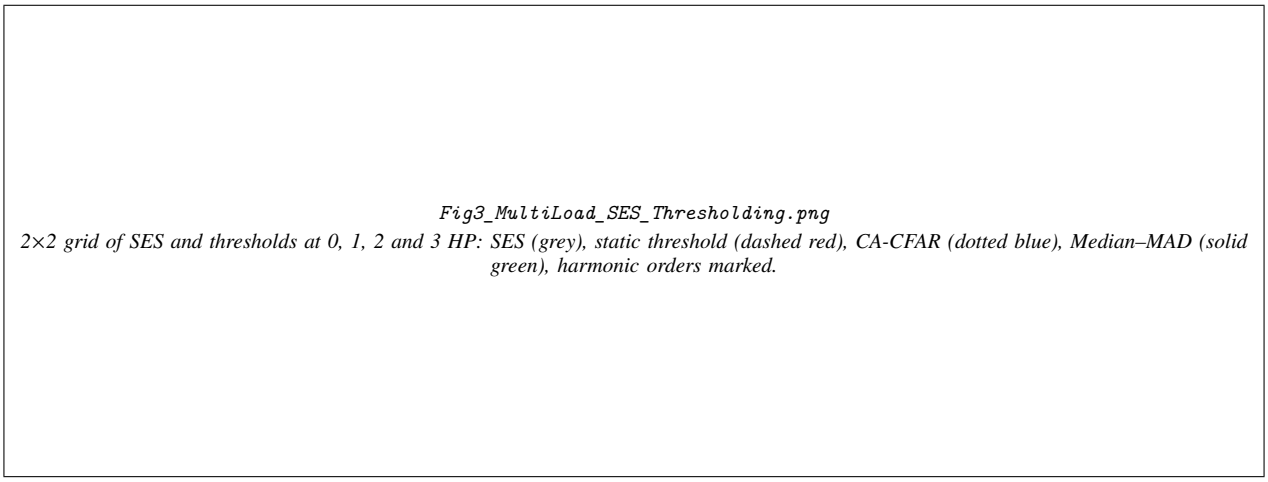


Fig. 3. Calibrated SES and comparative thresholding at four motor loads. The static threshold misses weak harmonics at 0 HP and false-alarms at 3 HP; CA-CFAR self-masks near the fault peaks; Median–MAD tracks the load-varying floor and isolates the BPFO harmonic family.

to a single classical result – the bilinear transform with pre-warping, the modulation theorem, and the order-statistic robustness of the median – so the detector contains no learned parameters and behaves predictably as the load changes.

The same structure suits embedded realisation. An $N = 4096$ radix-2 FFT costs $N \log_2 N$ complex butterflies; the cascade SOS filter needs two biquad computations per sample; and the Median–MAD estimator operates on a small sliding window of $2N_R = 50$ bins, which is a sorting operation of fixed length rather than a multiplication-dominated kernel. In 16-bit fixed-point arithmetic the expected signal-to-quantization-noise ratio of the datapath is $\mathrm{SQNR} \approx 6.02B + 1.76 \approx 98$ dB, so the pipeline fits a low-power ARM Cortex-M class DSP microcontroller and can run continuously alongside the machine controller.

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Author Contributions

Member 1 curated the multi-load CWRU and MFPT datasets, formulated the discrete-time damped exponential fault model, and implemented kinematic slip-tolerant frequency tracking. **Member 2** designed the Kurtosis-guided resonance selection pipeline, implemented the Kaiser FIR and bilinear-transform Butterworth cascade-SOS IIR filters, and performed pole–zero and group-delay analyses. **Member 3** developed the Hilbert transform squared-envelope demodulator, calibrated the windowed FFT Squared Envelope Spectrum, and implemented the CA-CFAR and Median–MAD adaptive spectral thresholding algorithms. **Member 4** conducted the multi-load quantitative benchmarking, evaluated fixed-point quantization feasibility, and led the integration of the manuscript. All authors reviewed and approved the final manuscript.

References

[1] R. B. Randall and J. Antoni, "Rolling element bearing diagnostics—A tutorial," *Mech. Syst. Signal Process.*, vol. 25, no. 2, pp. 485–520, 2011.
[2] S. Kim, D. An, and J.-H. Choi, "Diagnostics 101: A tutorial for fault diagnostics of rolling element bearing using envelope analysis in MATLAB," *Appl. Sci.*, vol. 10, no. 20, p. 7302, 2020.
[3] J. Antoni, "Fast computation of the kurtogram for the detection of transient faults," *Mech. Syst. Signal Process.*, vol. 21, no. 1, pp. 108–124, 2007.
[4] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms and Applications*, 4th ed. Upper Saddle River, NJ, USA: Pearson, 2006.
[5] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2009.
[6] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proc. IEEE*, vol. 66, no. 1, pp. 51–83, 1978.
[7] H. M. Finn and R. S. Johnson, "Adaptive detection mode with threshold control as a function of spatially sampled clutter-level estimates," *RCA Rev.*, vol. 29, pp. 414–464, 1968.
[8] P. J. Rousseeuw and C. Croux, "Alternatives to the median absolute deviation," *J. Amer. Statist. Assoc.*, vol. 88, no. 424, pp. 1273–1283, 1993.
[9] W. A. Smith and R. B. Randall, "Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study," *Mech. Syst. Signal Process.*, vol. 64–65, pp. 100–131, 2015.